Inner Product Spaces: Answer Key

Problem sheet for the Day 8 class · CSIR NET / GATE previous year questions

Answers

Q	1	2	3	4	5	6	7	8	9	10
Answer	(A)	(A)	(A)	(A)	(D)	2	(A),(B)	(B),(D) or (B),(C),(D)	(A),(B),(C)	(A),(B),(C)

Hints and short solutions

1. **(A)** GATE 2013 Skew-symmetric gives purely imaginary eigenvalues; orthogonal gives $|\lambda| = 1$. So $\lambda = \pm i$ (and 0 is impossible since the matrix is invertible).
2. **(A)** CSIR NET Dec 2016 Leading principal minors $3,\ 5,\ 12$ are all positive (eigenvalues $1, 3, 4$).
3. **(A)** GATE 2013 $\langle TX, Y\rangle = Y^t BAX$ and $\langle X, CY\rangle = Y^t C^t BX$. So $C^t B = BA$, i.e. $C = B^{-1}A^t B$.
4. **(A)** GATE 2015 $g - Pg$ is the projection of g on $V^{\perp}$: $\frac{\langle x^2, x\rangle}{\langle x, x\rangle}x = \frac{1/4}{1/3}x = \frac{3}{4}x$.
5. **(D)** GATE 2015 Eigenvalues λ of M are real, and $\lambda^2 \pm x\lambda + y$ has discriminant $x^2 - 4y < 0$, so it never vanishes.
6. **2** GATE 2015 The spanning vectors are orthonormal. $\langle v, w_1\rangle = \sqrt{2}$, $\langle v, w_2\rangle = 0$, so $d^2 = \|v\|^2 - 2 - 0 = 4 - 2 = 2$.
7. **(A),(B)** CSIR NET June 2016 Put $v = v_j$: $1 = 1 + \sum_{i\neq j}\langle v_i, v_j\rangle^2$, so all cross terms vanish. Orthonormal n vectors in $\mathbb{R}^n$ form a basis.
8. **(B),(D) or (B),(C),(D)** CSIR NET June 2016 $\dim V = n + 1$. A is the Gram matrix of the independent set $\{x^j\}$ for $\langle f, g\rangle = \int_0^1 fg$, so it is positive definite and $\det A > 0$. Option (C) prints $\mathbb{R}^n$ for $\mathbb{R}^{n+1}$, so the official key accepts both answers.
9. **(A),(B),(C)** CSIR NET Dec 2017 Eigenvalues are real and non-zero. Take p large for (A) and p even for (B),(C). For (D), a negative eigenvalue λ gives $e^{p\lambda} - 1 < 0$, e.g. $A = -I$.
10. **(A),(B),(C)** CSIR NET Dec 2017 AA^t is symmetric with quadratic form $\alpha\|x\|^2$, so $AA^t = \alpha I_m$. $A^t A$ shares the non-zero eigenvalues of AA^t: α (m times) and 0 ($n - m$ times).